On the Computation of Entropy, Reduced Units, and Two-Body Structural Contributions in Multicomponent Integral Equation Theories

HNCR-nCLJ Project Notes

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Abstract

This technical note details the statistical-mechanical formulation and physical interpretation of the entropy in multicomponent charged Lennard-Jones fluids solved via the Hypernetted Chain (HNC) integral equation. We address the occurrence of negative total entropy in classical continuous formulations using reduced Lennard-Jones units, explaining its connection to the thermal de Broglie wavelength, the Sackur–Tetrode equation, and the absence of an absolute zero in classical mechanics. Furthermore, we derive the exact multicomponent two-body pair entropy contribution $s^{(2)}$, outline its numerical implementation via Simpson quadrature, and analyze its relationship to the thermodynamic excess entropy s_{ex} and structural ordering.

1 Introduction

In classical liquid-state theory, the Ornstein–Zernike (OZ) equation with the Hypernetted Chain (HNC) closure provides a powerful, non-perturbative framework for determining the pair correlation functions $g_{ij}(r)$ and thermodynamics of dense neutral and ionic mixtures.

For a multicomponent fluid consisting of m distinct species with total number density $\rho = N/V$, particle mole fractions $x_i = N_i/N$, and partial densities $\rho_i = \rho x_i$, the thermodynamic excess properties are evaluated from the converged pair correlation functions $h_{ij}(r) = g_{ij}(r) - 1$ and direct correlation functions $c_{ij}(r)$ using Allnatt's renormalized OZ formulation.

The excess internal energy per particle u_{ex}/k_BT is computed from the radial pair potentials $u_{ij}(r)$ and correlation functions, while the excess Helmholtz free energy per particle a_{ex}/k_BT is evaluated via k-space integration over the direct and total correlation spectra. The thermodynamic excess entropy s_{ex} then follows directly from fundamental thermodynamics:

$$\frac{s_{\mathrm{ex}}}{Nk_B} = \frac{u_{\mathrm{ex}} - a_{\mathrm{ex}}}{Nk_BT}. \tag{1}$$

When combining s_{ex} with the ideal-gas mixture entropy s_{id} to obtain the total entropy $s = s_{\mathrm{id}} + s_{\mathrm{ex}}$, numerical calculations at low temperatures or high ionic strengths can yield negative values for s/Nk_B. As demonstrated below, this is a completely physical, well-understood characteristic of classical continuous statistical mechanics in reduced units.

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2 Classical vs. Quantum Entropy: The Third Law and Phase Space

In quantum statistical mechanics, entropy is defined via the density operator $\hat{\rho}$ as $S = -k_B\mathrm{Tr}(\hat{\rho}\ln\hat{\rho})$. By the Third Law of Thermodynamics (Nernst heat theorem), as temperature approaches ab-

solute zero, the system settles into its ground state (or a non-degenerate manifold), yielding:

$$\lim_{T\to 0} S_{\text{quantum}} = 0 \quad (\text{or } S \geq 0 \text{ strictly}). \tag{2}$$

In classical statistical mechanics, however, microstates form a continuous phase space $\Gamma = (\mathbf{r}^N, \mathbf{p}^N)$. The classical Gibbs entropy is an integral over the probability density $f(\Gamma)$:

$$S_{\text{classical}} = -k_B \int f(\Gamma) \ln[h^{3N} f(\Gamma)]\, d\Gamma. \tag{3}$$

Because $f(\Gamma)$ is a continuous probability density rather than a discrete probability, the Shannon/Gibbs continuous (differential) entropy has no absolute lower bound. Specifically, for an ideal gas of particles of mass m at temperature T, the momentum distribution is Gaussian with variance $\langle p^2 \rangle = mk_BT$. As $T \to 0$, the momentum distribution approaches a Dirac delta function in momentum space, causing the momentum-space volume to vanish. Consequently, the classical ideal-gas entropy diverges logarithmically to negative infinity:

$$\lim_{T\to 0} S_{\text{classical}} = -\infty. \tag{4}$$

Therefore, within classical mechanics, a negative value of entropy at lower temperatures is neither a mathematical anomaly nor a sign of unphysical code behavior; it is an inevitable consequence of continuous classical phase-space integration.

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3 Reduced Units and the Thermal de Broglie Wavelength

The exact quantum-corrected Sackur–Tetrode equation for a multicomponent ideal gas mixture of non-interacting particles is:

$$\frac{s_{\text{id}}}{Nk_B} = \frac{5}{2} - \sum_{i=1}^{m} x_i \ln\left(\rho_i \Lambda_i^3\right), \tag{5}$$

where Λ_i is the thermal de Broglie wavelength of species i:

$$\Lambda_i = \frac{h}{\sqrt{2\pi m_i k_B T}}. \tag{6}$$

Using the partial density relation $\rho_i = \rho x_i$ and expanding the logarithm:

$$\begin{aligned} \frac{s_{\text{id}}}{Nk_B} &= \frac{5}{2} - \ln\rho - \sum_{i=1}^{m} x_i \ln x_i - 3\sum_{i=1}^{m} x_i \ln \Lambda_i \\ &= \left[\frac{5}{2} - \ln(\rho\sigma^3) - \sum_{i=1}^{m} x_i \ln x_i\right] - 3\sum_{i=1}^{m} x_i \ln\left(\frac{\Lambda_i}{\sigma}\right), \end{aligned} \tag{7}$$

where σ is an arbitrary characteristic length scale (typically chosen as the Lennard-Jones collision diameter $\sigma = \sigma_{11} \equiv 1$).

3.1 Implicit Normalization in Reduced Units

In classical simulations and liquid-state integral equations, calculations are conducted in reduced Lennard-Jones units where $\sigma = 1$, $\epsilon = 1$, and $k_B = 1$. The ideal-gas baseline implemented in the code evaluates the bracketed term in Eq. (7):

$$\left(\frac{s_{\text{id}}}{Nk_B}\right)_{\text{reduced}} = \frac{5}{2} - \ln\rho^* - \sum_{i=1}^{m} x_i \ln x_i, \tag{8}$$

where $\rho^* = \rho\sigma^3$.

By adopting Eq. (8), the code implicitly sets $\Lambda_i = \sigma = 1$ (equivalent to setting Planck's constant and particle masses to a conventional reduced value).

3.2 Magnitude of the Omitted Quantum Phase-Space Offset

For actual physical atoms and ions in solution at ambient temperatures (e.g., argon, water, Na^+, Cl^- at $T \approx 300$ K):

- Characteristic atomic diameter: $\sigma \approx 3.0$–4.0 Å.
- Thermal de Broglie wavelength: $\Lambda \approx 0.15$–0.25 Å.

The ratio Λ/σ is therefore small: $\Lambda/\sigma \approx 0.04 \ll 1$. Evaluating the omitted logarithmic term:

$$-3 \ln\left(\frac{\Lambda}{\sigma}\right) \approx -3 \ln(0.04) \approx +9.66. \tag{9}$$

Thus, for any real physical fluid, the quantum phase-space constant shifts the baseline entropy upward by approximately $+8$ to $+12\, k_B$ per particle. When this realistic offset is restored, the absolute entropy of the physical liquid remains positive.

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4 Structural Ordering and Negative Excess Entropy ($s_{\mathbf{ex}} < 0$)

The excess entropy $s_{\text{ex}} = s - s_{\text{id}}$ quantifies the loss of accessible configurational volume caused by interparticle forces relative to an ideal gas at the same density and temperature:

- **Excluded-Volume Repulsion**: Steep repulsive cores prevent particle overlap, restricting particles to a fraction of the total system volume.
- **Coulombic Ordering**: Electrostatic attractions between counter-ions lead to charge alternation, local ion pairing, and solvation shells, imposing high spatial and orientational order.

Consequently, s_{ex}/Nk_B is strictly negative ($s_{\text{ex}} < 0$) for simple and ionic fluids, decreasing monotonically as density increases or temperature decreases (increasing $\beta^* = 1/T^*$).

For example, in a 5-component mixture at $\rho^* = 0.70$:

$$\left(\frac{s_{\text{id}}}{Nk_B}\right)_{\text{reduced}} = \frac{5}{2} - \ln(0.70) - \sum_{i=1}^{5} x_i \ln x_i \approx 2.50 + 0.3567 + 0.7410 \approx 3.60. \tag{10}$$

As β^* rises from 0.10 to 1.10, the structural ordering drives s_{ex}/Nk_B from -2.45 to -3.79. At $\beta^* \approx 0.90$, the magnitude of the structural ordering exceeds the reduced ideal baseline ($|s_{\text{ex}}| > s_{\text{id}}$), causing the total reduced entropy $s = s_{\text{id}} + s_{\text{ex}}$ to cross zero and become negative.

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5 The Two-Body Multiparticle Expansion ($s^{(2)}$)

5.1 Green–Wallace Multiparticle Expansion

The excess entropy can be rigorously expanded in an infinite series of multiparticle spatial correlation functions (Kirkwood, Green, Nettleton, and Wallace):

$$s_{\text{ex}} = s^{(2)} + s^{(3)} + s^{(4)} + \dots \tag{11}$$

where $s^{(n)}$ represents the contribution from n-particle irreducible correlations. In simple liquids and moderately coupled ionic fluids, the two-body contribution $s^{(2)}$ accounts for the vast majority (85%–95%) of the total excess entropy.

5.2 Multicomponent Formulation

For a mixture of m species with mole fractions x_i, the exact two-body entropy per particle is:

$$\frac{s^{(2)}}{Nk_B} = -\frac{1}{2}\rho \sum_{i=1}^{m}\sum_{j=1}^{m} x_i x_j \int \left[g_{ij}(\mathbf{r}) \ln g_{ij}(\mathbf{r}) - (g_{ij}(\mathbf{r}) - 1)\right] d\mathbf{r}. \tag{12}$$

For isotropic fluids, exploiting spherical symmetry ($d\mathbf{r} = 4\pi r^2 dr$) and the symmetry of the pair correlation functions ($g_{ij}(r) = g_{ji}(r)$):

$$\frac{s^{(2)}}{Nk_B} = -2\pi\rho \sum_{i=1}^{m}\sum_{j\le i} (2 - \delta_{ij})\, x_i x_j \int_0^\infty \left[g_{ij}(r) \ln g_{ij}(r) - h_{ij}(r)\right] r^2 dr, \tag{13}$$

where $h_{ij}(r) = g_{ij}(r) - 1$ is the total correlation function.

5.3 Mathematical Properties

1. **Negative Definiteness**: Consider the function $f(g) = g \ln g - g + 1$. By differentiation, $f'(g) = \ln g$ and $f''(g) = 1/g > 0$, so $f(g)$ is strictly convex with a unique global minimum at $g = 1$, where $f(1) = 0$. Hence:

$$g \ln g - (g - 1) \ge 0 \quad \forall g \ge 0. \tag{14}$$

   Because the integrand is non-negative everywhere, the two-body entropy is strictly negative:

$$\frac{s^{(2)}}{Nk_B} \le 0. \tag{15}$$

2. **Core Behavior**: Inside the repulsive core where $g_{ij}(r) \to 0$:

$$\lim_{g\to 0} g \ln g = 0 \implies [g \ln g - h] = -(-1) = 1. \tag{16}$$

   Integrating from $r = 0$ to $r = \sigma_{ij}$ yields an excluded-volume contribution of $-\frac{2\pi}{3}\rho\sigma_{ij}^3 x_i x_j$.

3. **Asymptotic Convergence**: At large separations ($r \to \infty$), $h(r) \to 0$. Expanding $g \ln g - h$:

$$(1 + h)\ln(1 + h) - h = \frac{1}{2}h^2 - \frac{1}{6}h^3 + \mathcal{O}(h^4). \tag{17}$$

   Because $h(r)$ decays faster than $1/r^2$ in screened ionic fluids, the integral converges rapidly and absolutely.

5.4 Numerical Integration in the HNCR Solver

In the modular solver, Eq. (13) is evaluated via Simpson's rule over the radial grid points $R_k = k \cdot \Delta r$:

$$I_{ij} = \int_0^{R_N} \left[g_{ij}(r) \ln g_{ij}(r) - h_{ij}(r)\right] r^2 dr \approx \frac{\Delta r^3}{3} + \frac{\Delta r}{3} \sum_{k=1}^{N-2} w_k R_k^2 f(g_{ij}(R_k)), \tag{18}$$

where $w_k \in \{1, 4, 2\}$ are standard Simpson weights, and the term $\Delta r^3/3$ accounts for the innermost interval $[0, \Delta r]$ where $g(r) = 0$.

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6 Results and Thermodynamic Consistency

Table 1 summarizes the results for a 5-component charged Lennard-Jones mixture at total density $\rho^* = 0.70$ across a wide range of coupling parameters $\beta^* = 1/T^* \in [0.10, 1.10]$.

The 5-component system comprises:

- Species 1: Neutral solvent ($q_1 = 0$, $x_1 = 0.80$, $\sigma_{11} = 1.0$, $\epsilon_{11} = 1.0$).
- Species 2: Monovalent cation ($q_2 = +1$, $x_2 = 0.05$).
- Species 3: Divalent cation ($q_3 = +2$, $x_3 = 0.04$).
- Species 4: Monovalent anion ($q_4 = -1$, $x_4 = 0.09$).
- Species 5: Divalent anion ($q_5 = -2$, $x_5 = 0.02$).

Table 1: Thermodynamic properties and entropy components of the 5-component charged LJ mixture at $\rho^* = 0.70$.

β^*	Z_{virial}	Z_a	u_{ex}/k_BT	a_{ex}/k_BT	s_{ex}/Nk_B	s_{tot}/Nk_B	$s^{(2)}/Nk_B$	μ_1/k_BT	χ_T^{-1}
0.10	3.2151	3.2208	0.0091	2.4568	-2.4478	1.1609	-0.9800	4.6855	6.0784
0.15	3.3828	3.3892	-0.1822	2.4250	-2.6072	1.0015	-1.0960	4.8278	6.5021
0.20	3.4639	3.4708	-0.3883	2.3446	-2.7329	0.8757	-1.1897	4.8353	6.7295
0.30	3.4801	3.4878	-0.8249	2.1055	-2.9304	0.6783	-1.3412	4.6268	6.8551
0.40	3.3800	3.3882	-1.2807	1.8061	-3.0868	0.5219	-1.4657	4.2429	6.7199
0.50	3.2076	3.2163	-1.7486	1.4702	-3.2188	0.3898	-1.5741	3.7511	6.4234
0.70	2.7264	2.7359	-2.7084	0.7302	-3.4386	0.1701	-1.7617	2.5644	5.5268
0.90	2.1317	2.1418	-3.6910	-0.0683	-3.6227	-0.0140	-1.9260	1.2075	4.3781
1.10	1.4705	1.4810	-4.6926	-0.9059	-3.7868	-0.1781	-2.0776	-0.2535	3.0816

6.1 Key Observations

1. **Thermodynamic Consistency**: The virial compressibility factor Z_{virial} and the free-energy / chemical-potential compressibility factor $Z_a = 1 + \sum_i x_i\mu_i/k_BT - a_{\text{ex}}/k_BT$ agree to within $< 0.5\%$ across the entire temperature span, verifying that the numerical HNC solution is sound and highly accurate.

2. **Monotonic Entropy Decline**: Both s_{ex} and $s^{(2)}$ decrease monotonically as β^* increases (temperature drops), reflecting enhanced spatial structuring and ion pairing.

3. **Transition to Negative Total Entropy**: The total reduced entropy s_{tot} smoothly crosses zero at $\beta^* \approx 0.89$ and becomes -0.1781 at $\beta^* = 1.10$. As established in Section 3, adding the physical quantum offset $-3\ln(\Lambda/\sigma) \approx +9.66$ shifts the true entropy of the physical fluid to $s_{\text{real}}/Nk_B \approx +9.48 > 0$.

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7 Summary

- In classical statistical mechanics, continuous phase-space integration leads to $\lim_{T\to 0} S = -\infty$, meaning that negative values of entropy at low temperatures are mathematically natural and expected.

- Reduced units implicitly set the thermal de Broglie wavelength to unity ($\Lambda = \sigma = 1$), omitting a large positive quantum phase-space term (typically $+8$–$12\,k_B$ per particle for physical fluids).
- The excess entropy s_{ex} and its two-body pair approximation $s^{(2)}$ are non-positive definite, mass-independent quantities that directly characterize the degree of structural ordering and govern dynamical transport coefficients via Rosenfeld scaling.